

Expt-9: Field Characteristics of Helmholtz Coil

Kalamata Chaitanya

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1 Aim of the Experiment

To study the variation of the magnetic field with distance along the axis of a circular coil carrying current, and the principle of superposition of magnetic fields.

2 Apparatus Used

1. Helmholtz Coil Setup
2. Magnetic Field Measurement Unit

The setup includes **two circular coils** and a **Hall-effect sensor** connected to a digital Gauss meter for measuring the magnetic field.

3 Theory Used

When an electric current flows through a circular coil, it produces a magnetic field around it. Along the axis of the coil, the magnetic field is **maximum at the centre** of the coil and gradually decreases as the distance from the centre increases on either side.

By measuring the magnetic field at different positions along the axis, the variation of the magnetic field with distance can be studied.

The experiment also uses two circular coils. When each coil carries current separately, each produces its own magnetic field. When both coils carry current simultaneously, the resultant magnetic field at any point is equal to the sum of the magnetic fields produced by the two coils individually. This demonstrates the **principle of superposition of magnetic fields**.

When the separation between the two coils is equal to their radius, the magnetic field in the region between the coils becomes nearly uniform. This arrangement is known as the **Helmholtz coil configuration** and is widely used where a uniform magnetic field is required.

4 Formula Used

For a circular coil of:

- radius a ,
- number of turns n ,
- carrying current I ,

the magnetic field at a point on its axis, at a distance x from its centre, is

$$B(x) = \frac{\mu_0 n I a^2}{2(a^2 + x^2)^{3/2}}$$

where:

- B = magnetic field at the point,
- μ_0 = permeability of free space,
- n = number of turns,
- I = current through the coil,
- a = radius of the coil,
- x = distance from the centre along the axis.

The field is maximum at the centre of the coil, where $x = 0$.

5 Observations

Position of Coil-1 = 0 mm

Position of Coil-2 = 112 mm

Distance between coils = 112 mm

Current in the coils = 500 mA

Table 1: Observation Table: Magnetic Field Measurements

S. No.	Sensor Stand Position (mm)	Magnetic Field (gauss)		
		Coil-1	Coil-2	Both Coils
1	205	2.4	-5.1	4.5
2	210	2.8	-5.0	5.1
3	215	3.3	-4.9	5.8
4	220	3.9	-4.6	6.5
5	225	4.5	-4.4	7.2
6	230	5.0	-4.2	8.0
7	235	5.4	-4.0	8.6
8	240	5.7	-3.8	9.1
9	245	6.1	-3.6	9.7
10	250	6.3	-3.4	10.1

Table 1: Observation Table: Magnetic Field Measurements

S. No.	Sensor Stand Position (mm)	Magnetic Field (gauss)		
		Coil-1	Coil-2	Both Coils
11	255	6.5	-3.1	10.7
12	260	6.8	-2.8	11.2
13	265	6.8	-2.5	11.6
14	270	6.8	-2.1	11.8
15	275	6.7	-1.7	12.0
16	280	6.4	-1.3	12.2
17	285	6.1	-0.9	12.3
18	290	5.8	-0.5	12.3
19	295	5.4	-0.2	12.3
20	300	4.9	0.3	12.4
21	305	4.4	0.7	12.4
22	310	3.9	1.2	12.4
23	315	3.3	1.7	12.3
24	320	2.8	2.3	12.3
25	325	2.3	2.8	12.3
26	330	1.7	3.3	12.3
27	335	1.2	3.9	12.3
28	340	0.7	4.5	12.3
29	345	0.2	4.9	12.3
30	350	-0.2	5.3	12.3
31	355	-0.6	5.7	12.3
32	360	-1.1	6.2	12.3
33	365	-1.5	6.3	12.0
34	370	-1.8	6.5	11.9
35	375	-2.1	6.6	11.7
36	380	-2.5	6.7	11.4
37	385	-2.8	6.6	11.0
38	390	-3.1	6.5	10.6
39	395	-3.4	6.3	10.1
40	400	-3.6	6.1	9.8
41	405	-3.8	5.8	9.2
42	410	-4.0	5.4	8.6
43	415	-4.2	4.9	7.9
44	420	-4.4	4.6	7.3
45	425	-4.5	4.1	6.7
46	430	-4.7	3.5	5.9
47	435	-4.9	2.8	5.1
48	440	-5.0	2.3	4.5
49	445	-5.1	1.8	3.9
50	450	-5.3	1.2	3.2

Table 1: Observation Table: Magnetic Field Measurements

S. No.	Sensor Stand Position (mm)	Magnetic Field (gauss)		
		Coil-1	Coil-2	Both Coils
51	455	-5.4	0.8	2.6
52	460	-5.5	0.4	2.1
53	465	-5.5	0.0	1.5
54	470	-5.6	-0.5	0.9
55	475	-5.7	-0.9	0.4
56	480	-5.8	-1.3	0.0
57	485	-5.9	-1.7	-0.4
58	490	-6.0	-2.1	-0.8
59	495	-6.0	-2.4	-1.3
60	500	-6.1	-2.7	-1.6

6 Analysis on observations

For Coil-1:

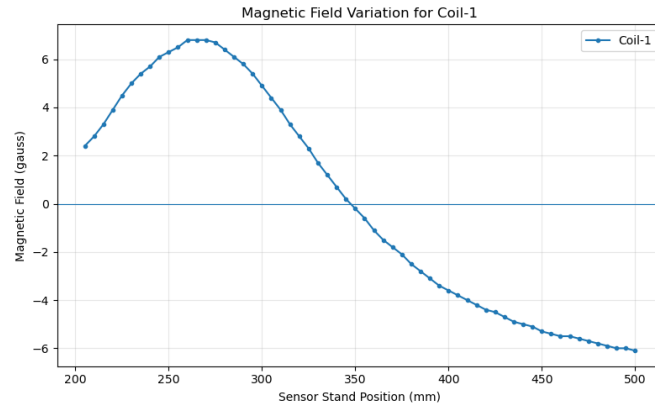


Figure 1: Magnetic Field (gauss) vs Sensor Stand Position (mm) for Coil-1

For Coil-2:

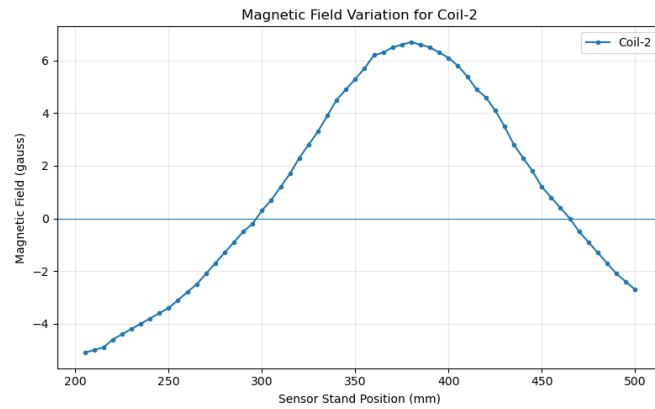


Figure 2: Magnetic Field (gauss) vs Sensor Stand Position (mm) for Coil-2

For both coils:

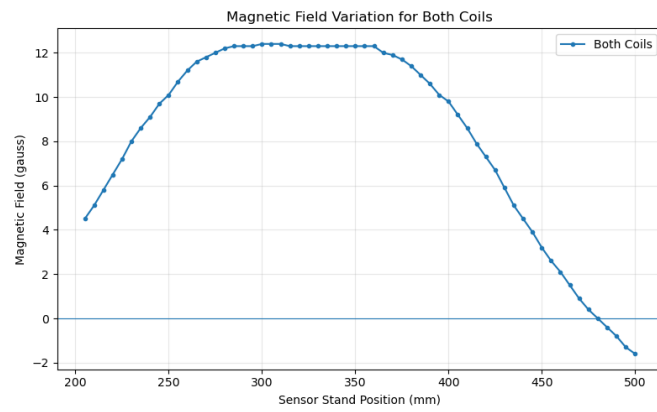


Figure 3: Magnetic Field (gauss) vs Sensor Stand Position (mm) for both coils

Plotting every graph on a single graph:

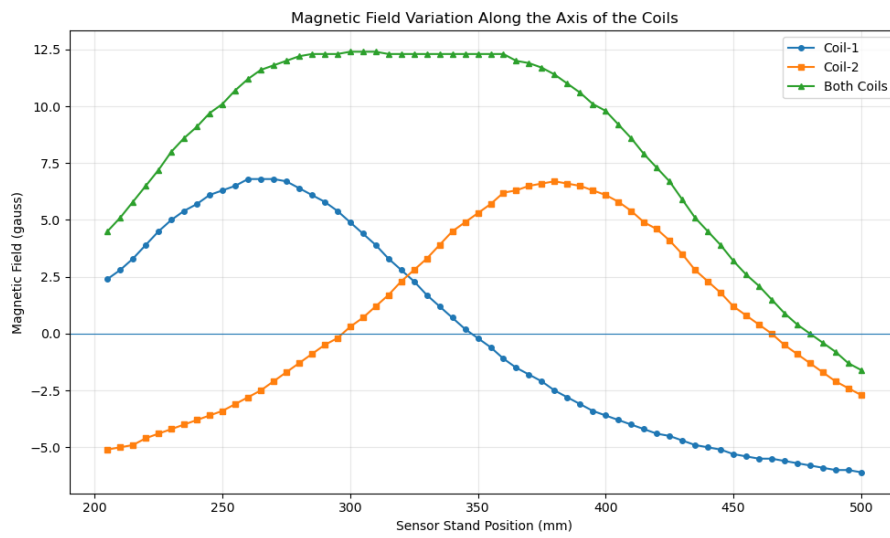


Figure 4: Magnetic Field (gauss) vs Sensor Stand Position (mm)

7 Error Analysis

Comparing my observational values with theory:

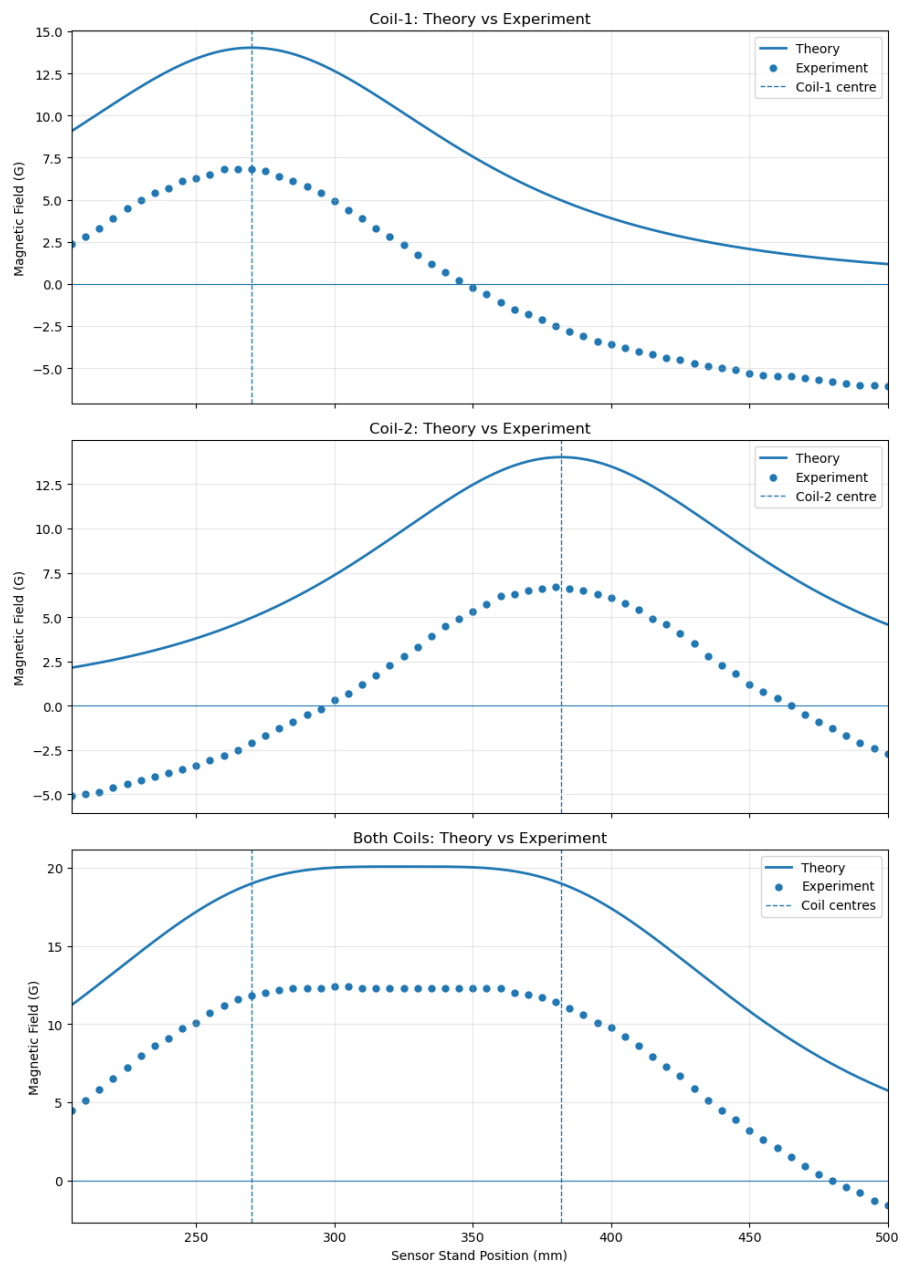


Figure 5: Theory vs Experiment

Calculating residuals using

$$R(x) = B_{exp}(x) - B_{theory}(x)$$

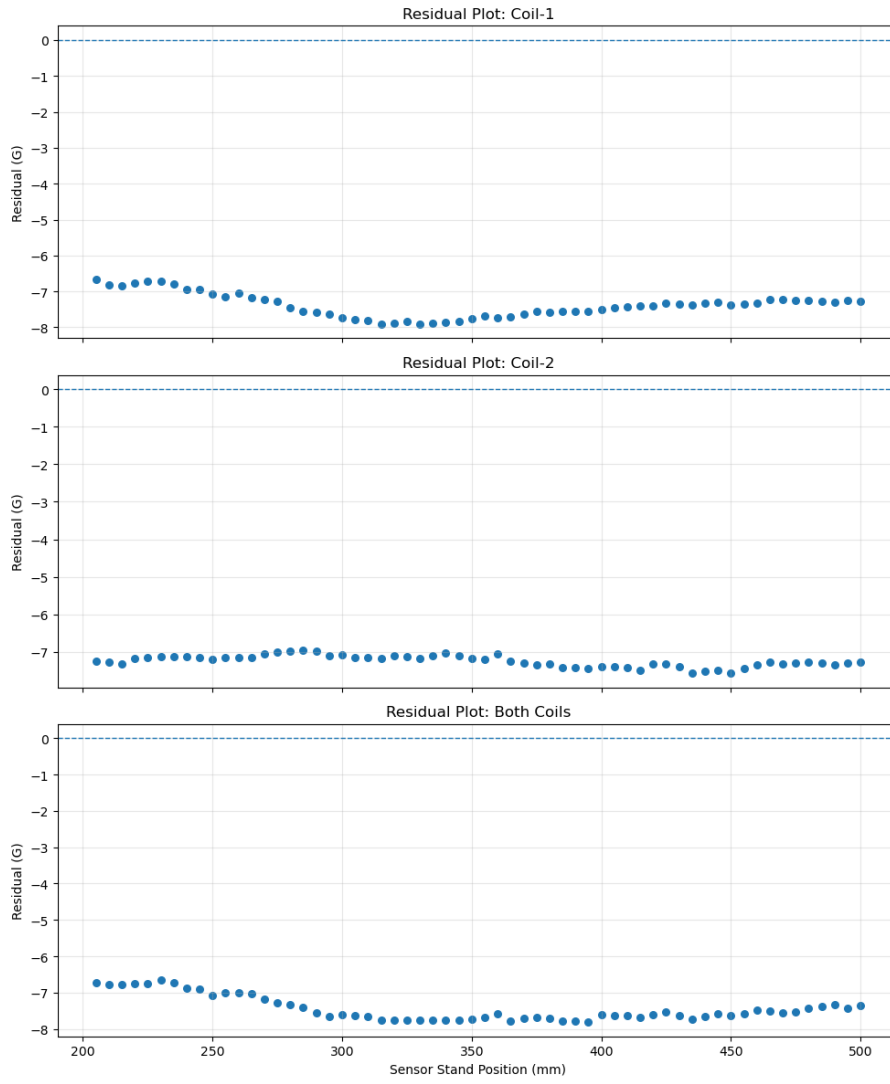


Figure 6: Residual Plots